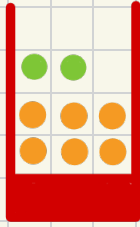




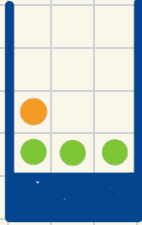
# Probability theory

Red Box (r)



$$p(B=r) = \frac{4}{10}$$

Blue Box (b)



$$p(B=b) = \frac{6}{10}$$

a) identify all random variables in this problem and their possible values

$B(\text{box})$ : Can take values  $\{r, b\}$  for red and blue boxes.

$F(\text{fruit})$ : Can take values  $\{a, o\}$  for apple and oranges.

## Basic probabilities

a)  $P(B=r) = \frac{4}{10}$     b)  $P(B=b) = 1 - P(B=r) = 1 - \frac{4}{10} = \frac{6}{10}$     c)  $P(F=a | B=r) = \frac{2}{4} = \frac{1}{2}$

d)  $P(F=o | B=r) = \frac{6}{4} = \frac{3}{2}$     e)  $P(F=a | B=b) = \frac{1}{6}$     f)  $P(F=o | B=b) = \frac{1}{2}$

## Marginal probabilities

- we first setup the joint probability table by using the Product rule:  $p(X,Y) = p(Y|X)p(X)$

|     | B=r                          | B=b                          |
|-----|------------------------------|------------------------------|
| F=a | $P(B=r, F=a) = \frac{1}{10}$ | $P(B=b, F=a) = \frac{1}{20}$ |
| F=o | $P(B=r, F=o) = \frac{3}{10}$ | $P(B=b, F=o) = \frac{3}{20}$ |

$$\begin{aligned}
 P(B=r, F=a) &= P(B=r) \cdot P(F=a|B=r) = \frac{4}{10} \cdot \frac{1}{2} = \frac{4}{20} = \frac{1}{5} \\
 P(B=r, F=o) &= P(B=r) \cdot P(F=o|B=r) = \frac{4}{10} \cdot \frac{3}{4} = \frac{12}{40} = \frac{3}{10} \\
 P(B=b, F=a) &= P(B=b) \cdot P(F=a|B=b) = \frac{6}{10} \cdot \frac{1}{6} = \frac{1}{10} \\
 P(B=b, F=o) &= P(B=b) \cdot P(F=o|B=b) = \frac{6}{10} \cdot \frac{1}{2} = \frac{3}{10}
 \end{aligned}$$

Summing the rows:  $\frac{1}{10} + \frac{3}{10} = \frac{4}{10}$  and  $\frac{1}{20} + \frac{3}{20} = \frac{4}{20} = \frac{1}{5}$ .  
Summing the columns:  $\frac{1}{10} + \frac{1}{20} = \frac{3}{20}$  and  $\frac{3}{10} + \frac{3}{20} = \frac{9}{20}$ .  
Total sum:  $\frac{4}{10} + \frac{1}{5} = \frac{4}{10} + \frac{2}{10} = \frac{6}{10} = \frac{3}{5}$ .  
Wait, the total sum should be 1. Let's recalculate:  $\frac{1}{10} + \frac{3}{10} = \frac{4}{10}$  and  $\frac{1}{20} + \frac{3}{20} = \frac{4}{20} = \frac{1}{5}$ . Total is  $\frac{4}{10} + \frac{1}{5} = \frac{4}{10} + \frac{2}{10} = \frac{6}{10} = \frac{3}{5}$ . This is incorrect. The correct sum is  $\frac{1}{10} + \frac{3}{10} + \frac{1}{20} + \frac{3}{20} = \frac{2}{20} + \frac{6}{20} + \frac{1}{20} + \frac{3}{20} = \frac{12}{20} = \frac{3}{5}$ . The table values are correct, but the marginal sums are wrong. The correct marginal sums are  $\frac{4}{10}$  for F=a and  $\frac{1}{5}$  for F=o, and  $\frac{3}{10}$  for B=r and  $\frac{9}{20}$  for B=b. The total sum is  $\frac{4}{10} + \frac{1}{5} = \frac{6}{10} = \frac{3}{5}$ . This is a mistake in the original image. The correct total sum is 1.

a)  $P(F=a) = \sum_{B=i} P(F=a|B=i)$

$$= \frac{1}{10} + \frac{1}{20} = \frac{2+1}{20} = \frac{3}{20}$$

b)  $P(F=o) = \sum_{B=i} P(F=o|B=i) = \frac{3}{10} + \frac{1}{20} = \frac{6+1}{20} = \frac{7}{20}$

c) does  $P(F=a) + P(F=o) = ?$

$$P(F=a) + P(F=o) = \frac{3}{20} + \frac{7}{20} = \frac{10}{20} = \frac{1}{2}$$

The sum rule  $P(x) = \sum_y p(X,Y)$

a) Derive Bayes thm. using the product rule

$$\begin{aligned}\text{Product rule: } p(X, Y) &= p(Y|X) p(X) \\ &= p(X|Y) p(Y) \\ \Rightarrow p(X|Y) p(Y) &= p(Y|X) p(X) \\ \Rightarrow p(X|Y) &= \frac{p(Y|X) p(X)}{p(Y)}\end{aligned}$$

b) Calculate  $P(B=r | F=o)$

$$P(B=r | F=o) = \frac{P(F=o|B=r) P(B=r)}{P(F=o)} = \frac{\frac{3}{10} \cdot \frac{4}{10}}{\frac{9}{20}} = \frac{1.5}{2} = \frac{2}{3}$$

c) Calculate  $P(B=b | F=o) = 1 - \frac{2}{3} = \frac{1}{3}$

$$\begin{aligned}\text{d) Calculate } P(B=r | F=a) &= \frac{P(F=a|B=r) P(B=r)}{P(F=a)} \\ &= \frac{\frac{1}{4} \cdot \frac{4}{10}}{\frac{11}{20}} = \frac{2}{11}\end{aligned}$$

e) Calculate  $P(B=b | F=a) = 1 - \frac{2}{11} = \frac{9}{11}$

**Notice:** Observing an orange shifts our belief toward the Red box from prior 0.4 to posterior 0.67

Observing an apple shifts our belief toward the blue box from prior 0.6 to posterior 0.82

# Sum & Product Rule

a) we know  $P(\text{Rain}|\text{Cloudy})$  &  $P(\text{Cloudy})$ . Find  $P(\text{Rain}, \text{Cloudy})$

$$P(\text{Rain}, \text{Cloudy}) = P(\text{Rain}|\text{Cloudy}) P(\text{Cloudy})$$

- we are converting from conditional probability  $P(Y|X)$  to joint  $P(Y, X)$ . The chain rule allows us to factorize joint distributions to products of conditional and marginal.

b) Know  $P(\text{Pass}, \text{Math})$  &  $P(\text{Pass}, \text{Science})$ . Find  $P(\text{Pass})$

$$P(\text{Pass}) = \sum_{\text{Subject}_i} P(\text{Pass}, \text{Subject}_i)$$

c) Know  $P(A|B)$ ,  $P(A|B')$ ,  $P(B)$ . Find  $P(A)$

- event  $B$  partitions the sample space to  $\{B, B'\}$ . So  $A = (A \cap B) \cup (A \cap B')$

- since these are disjoint  $P(A) = P(A \cap B) + P(A \cap B')$

- using conditional probability we define  $P(A \cap B) = P(A|B) P(B)$

$$\therefore \text{The law of total probability } P(A) = P(A|B) P(B) + P(A|B') P(B')$$

$$P(\text{Sunny}) = 1 - P(\text{Cloudy}) = 0.6$$

$$P(\text{Cloudy}) = 0.4$$

$$P(\text{Rain}|\text{Cloudy}) = 0.7$$

$$P(\text{Rain}|\text{Sunny}) = 0.1$$

Find  $P(\text{Rain}, \text{Cloudy})$ ,  $P(\text{Rain})$

$$P(\text{Rain}, \text{Cloudy}) = P(\text{Rain}|\text{Cloudy}) \cdot P(\text{Cloudy}) = 0.7 \cdot 0.4 = 0.28$$

$$\begin{aligned} P(\text{Rain}) &= P(\text{Rain}, \text{Cloudy}) + P(\text{Rain}, \text{Sunny}) \\ &= P(\text{Rain}|\text{Cloudy}) \cdot P(\text{Cloudy}) + P(\text{Rain}|\text{Sunny}) \cdot P(\text{Sunny}) \\ &= 0.28 + 0.06 \end{aligned}$$

• 2% of population  $\oplus$

• 98% of pop  $\ominus$

• Test finds 95%  $\oplus$

• Test finds 98% of  $\ominus$

• Test finds 2% of  $\oplus$

• Test finds 5% of  $\oplus$

Find  $P(\oplus \text{ test}, \oplus)$

$$P(\oplus \text{ test}, \oplus) = 0.95 \cdot 0.02 = 0.019$$

$$\text{Find } P(\text{Positive test}) = 0.95 \cdot 0.02 + 0.02 \cdot 0.98$$

$$= 0.019 + 0.0196$$

$$= 0.0386$$

$$P(\text{Disease}|\text{Positive}) = \frac{P(\text{Positive}|\text{Disease}) \cdot P(\text{Disease})}{P(\text{Positive})}$$

$$= \frac{0.95 \cdot 0.02}{0.0386} = 0.49$$

A factory has 3 production lines with different characteristics

| Line | $P(\text{line})$ | $P(\text{Defect} \text{Line})$ |
|------|------------------|--------------------------------|
| A    | 0,5              | 0,02                           |
| B    | 0,3              | 0,05                           |
| C    | 0,2              | 0,08                           |

$$\begin{aligned} \text{a) } P(\text{Defect}, \text{Line A}) &= P(\text{Defect}|\text{Line}) P(\text{Line A}) \\ &= 0,5 \cdot 0,02 \\ &= \underline{\underline{0,01}} \end{aligned}$$

$$\begin{aligned} \text{b) } P(\text{defect}) &= P(\text{Defect}, \text{A}) + P(\text{Defect}, \text{B}) + P(\text{Defect}, \text{C}) \\ &= P(\text{Defect}|\text{Line}) P(\text{Line A}) + P(\text{Defect}|\text{Line}) P(\text{Line B}) + P(\text{Defect}|\text{Line}) P(\text{Line C}) \\ &= 0,01 + 0,015 + 0,016 \\ &= \underline{\underline{0,041}} \end{aligned}$$

c) If a defective item is found, what is a probability it is from line C?

$$\begin{aligned} P(\text{Line C}|\text{Defect}) &= \frac{P(\text{Defect}|\text{Line C}) \cdot P(\text{Line C})}{P(\text{Defect})} \\ &= \frac{0,08 \cdot 0,2}{0,041} \\ &= \underline{\underline{0,39}} \end{aligned}$$

Products go through two quality checks

- First check:  $P(\text{Pass}_1) = 0.9$
- Second check:  $P(\text{Pass}_2 | \text{Pass}_1) = 0.95$ ,  $P(\text{Pass}_2 | \text{Fail}_1) = 0.3$

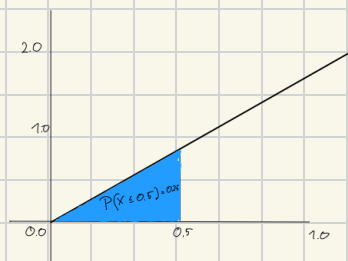
$$\begin{aligned} \text{a) } P(\text{Pass}_1, \text{Pass}_2) &= P(\text{Pass}_2 | \text{Pass}_1) \cdot P(\text{Pass}_1) \\ &= \underline{\underline{0.855}} \end{aligned}$$

$$\begin{aligned} \text{b) } P(\text{Pass}_2) &= P(\text{Pass}_2 | \text{Pass}_1) \cdot P(\text{Pass}_1) + P(\text{Pass}_2 | \text{Fail}_1) \cdot P(\text{Fail}_1) \\ &= 0.855 + P(\text{Pass}_2 | \text{Fail}_1) \cdot P(\text{Fail}_1) \\ &= 0.855 + P(\text{Pass}_2 | \text{Fail}_1) \cdot (1 - P(\text{Pass}_1)) \\ &= 0.855 + 0.03 \\ &= \underline{\underline{0.885}} \end{aligned}$$

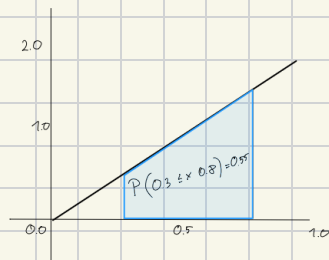
Given triangular distribution  $p(x) = 2x \forall x \in [0, 1]$

a) find  $P(X \leq 0.5)$

$$P(X \leq 0.5) = \int_0^{0.5} 2x \, dx = [x^2]_0^{0.5} = 0.5^2 = 0.25$$



b) find  $P(0.3 \leq X \leq 0.8) = \int_{0.3}^{0.8} 2x \, dx = [x^2]_{0.3}^{0.8} = 0.8^2 - 0.3^2 = \underline{\underline{0.55}}$



|                                                     |  |
|-----------------------------------------------------|--|
| Expectation $E[f(x)]$                               |  |
| • discrete $E[f(x)] = \sum_x p(x) f(x)$             |  |
| • Continuous $E[f(x)] = \int p(x) f(x) dx$          |  |
| • Sampling $E[f] = \frac{1}{n} \sum_{i=1}^n f(x_i)$ |  |

|                                       |  |
|---------------------------------------|--|
| Variance $V[x]$                       |  |
| def $Var[X] = E[(X - E[X])^2]$        |  |
| alt def: $Var[X] = E[X^2] - (E[X])^2$ |  |
| properties: $Var[aX+b] = a^2 Var[X]$  |  |

|                                                         |  |
|---------------------------------------------------------|--|
| Covariance $Cov[X, Y]$                                  |  |
| def: $Cov[X, Y] = E[(X - E[X])(Y - E[Y])]$              |  |
| alt def: $Cov[X, Y] = E[XY] - E[X]E[Y]$                 |  |
| independence: if $X, Y \perp \Rightarrow Cov[X, Y] = 0$ |  |

Bernoulli distribution

$$P(X=x) = p^x (1-p)^{1-x} \quad \forall x \in \{0,1\}$$

- find  $E[X]$  &  $Var[X]$ ,  $P(X=1)=p$ ,  $P(X=0)=1-p$

Solution

$$\begin{aligned} E[X] &= \sum_{x=0}^1 x P(X=x) \\ &= 0 \cdot \underbrace{P(X=0)}_{1-p} + 1 \cdot \underbrace{P(X=1)}_p \\ &= 0 \cdot (1-p) + p \\ &= p \end{aligned}$$

$$\begin{aligned} E[X^2] &= \sum_{x=0}^1 x^2 P(X=x) \\ &= 0^2 P(X=0) + 1^2 P(X=1) \\ &= 0^2 (1-p) + 1^2 p \\ &= p \end{aligned}$$

$$\begin{aligned} Var[X] &= E[X^2] - (E[X])^2 \\ &= p - p^2 \\ &= \underline{p(1-p)} \end{aligned}$$

Continuous Expectations

a) Uniform distribution on  $[a,b]$ :  $p(x) = \frac{1}{(b-a)}$  Find  $E[X]$ ,  $V[X]$ .

$$\begin{aligned} E[X] &= \int_a^b x p(x) dx \\ &= \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x dx = \frac{1}{b-a} \left[ \frac{x^2}{2} \right]_a^b = \frac{1}{b-a} \left[ \frac{b^2}{2} - \frac{a^2}{2} \right] = \frac{(b^2 - a^2)}{2(b-a)} = \frac{(b+a)}{2} \end{aligned}$$

$$\begin{aligned} E[X^2] &= \int_a^b x^2 p(x) dx \\ &= \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x^2 dx = \frac{1}{b-a} \left[ \frac{x^3}{3} \right]_a^b = \frac{1}{b-a} \left[ \frac{b^3}{3} - \frac{a^3}{3} \right] = \frac{1}{(b-a)} \cdot \frac{(b^3 - a^3)}{3} = \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} = \frac{b^2 + ab + a^2}{3} \end{aligned}$$

$$V[X] = E[X^2] - \frac{(a^2 + ab + a^2)}{3} - \left( \frac{a+b}{2} \right)^2 = \frac{(b-a)^2}{12}$$

Properties of Variance

a) if  $X \sim U[0,10]$ , find  $Var[3X-5]$

$$Var[3X-5] = 9 Var[X] = 9 \cdot \frac{(10-0)^2}{12} = 75$$

b) If  $Y = aX+b$ , where  $\mu$  and  $\sigma^2$  find  $E[Y]$  &  $V[Y]$

$$\begin{aligned} E[Y] &= a E[X] + b & V[Y] &= a^2 V[X] \\ &= a\mu + b & &= a^2 \sigma^2 \end{aligned}$$

# Bayesian Theorem

$$P(\text{Parameters} | \text{data}) = \frac{P(\text{data} | \text{parameters}) P(\text{parameters})}{P(\text{data})}$$

and usually  $\text{posterior} \propto \text{likelihood} \times \text{Prior}$

## Exercise

a coin is tossed 3x, resulting in 3x+1.

- step 1: Prior - before we observe data, we believe any probability of heads between [0,1] is equally likely.

$$p \sim \text{Beta}(1,1)$$

- step 2: Likelihood - how we incorporate our data: we observed 3 heads in 3 trials. The likelihood is the probability of observing the data given a specific  $p$  is  $p^3$ :

$$\text{Likelihood} \propto p^3$$

- the coin has prob  $p$  of heads, the prob. of getting 3 heads in a row is  $p \cdot p \cdot p = p^3$

step 3: Posterior distribution

- how we combine the prior  $\wedge$  likelihood

• Prior:  $\text{Beta}(1,1)$

• Likelihood:  $p^3$

• Posterior:  $\text{Beta}(1+3, 1+0) = \text{Beta}(4,1)$

↳ Beta is conjugate to Binomial likelihood

↳ posterior is also Beta distribution and we update the parameters in a following way.

$$\alpha_{\text{posterior}} = \alpha_{\text{prior}} + \text{number of heads} = 1+3 = 4$$

$$\beta_{\text{posterior}} = \beta_{\text{prior}} + \text{num. of tails} = 1+0 = 1$$

step 4: posterior mean + intervals + variance

$$E[p] = \frac{\alpha}{\alpha + \beta}$$

$$V[p] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$\text{mean of } p: E[p | \text{data}] = \frac{4}{4+1} = 0.8$$

probability that  $p > 0.5$

$$P(p > 0.5 | \text{data}) = \int_{0.5}^1 4p^3 dp$$

- Beta(4,1) has PDF:  $f(p) = 4p^3$

- so we integrate

$$\int_{0.5}^1 4p^3 dp = \left[ p^4 \right]_{0.5}^1 = 1^4 - (0.5)^4 = 1 - \frac{1}{16} = \frac{15}{16} = 0.9375$$



- what would different likelihoods look like?

- we again model coin flips

• Binomial Likelihood  $\text{Bin}(h|n, p) \propto p^h (1-p)^{n-h}$

• Posterior Beta:  $p \sim \text{Beta}(1+h, 1+n-h)$

ex. 2 heads out of 3 trials,  $n=3$ ,  $h=2$

$\therefore$  likelihood  $p^h (1-p)^{n-h} = \underline{\underline{p^2 (1-p)}}$   $\therefore$  posterior Beta(1+2, 1+1)

# Probabilistic curve fitting

- assumption: target values have  $\epsilon \sim \mathcal{N}(0, \beta)$  around the true function:  $t = y(x, w) + \epsilon$

•  $\beta$  is a precision parameter (inverse variance)

•  $\hat{t} = y(x, w)$  - deterministic function

$\Rightarrow$  which gives:  $p(\hat{t} | x, w, \beta) = \mathcal{N}(\hat{t} | y(x, w), \beta^{-1})$

*Annotations: "input features" points to  $x$ , "model weights" points to  $w$ , "linear model output" points to  $y(x, w)$ , "precision of  $\epsilon$  noise" points to  $\beta$ .*

## Example

$x = [1 \ 2]^T$ ;  $w = [0.5 \ -1]^T$ ;  $\beta = 4 \Rightarrow \sigma^2 = \frac{1}{\beta} = 0.25$ ;  $\hat{t} = -1.0$

a) calculate model output

$$y(x, w) = w^T x = [0.5 \ -1] \begin{bmatrix} 1 \\ 2 \end{bmatrix} = (0.5 \cdot 1) + (-1 \cdot 2) = 0.5 - 2 = -1.5$$

$$f(x, w) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-w)^2}{2\sigma^2}}$$

b) we use Gaussian PDF and assume

$$\epsilon \sim \mathcal{N}(-1.5, 0.25)$$

the likelihood is:  $p(\epsilon | x, w, \beta) = \frac{1}{\sqrt{2\pi \cdot 0.25}} \exp\left(-\frac{(\epsilon - (-1.5))^2}{2 \cdot 0.25}\right)$

$$\begin{aligned} \rightarrow \text{plugging in } \hat{t} = -1.0 &= \frac{1}{\sqrt{0.5\pi}} \exp\left(-\frac{(0.5)^2}{0.5}\right) \\ &= \frac{1}{\sqrt{0.5\pi}} \exp(-0.5) \\ &\approx \underline{\underline{0.4839}} \end{aligned}$$

Given  $p(\hat{t} | x, w, \beta) = \mathcal{N}(\hat{t} | y(x, w), \beta^{-1}) \rightarrow$  Given input  $x$  a parameters  $w$  and  $\beta$ , the target value  $\hat{t}$  follows a Gaussian distribution w/  $\mu = y(x, w)$  and  $\sigma^2 = \frac{1}{\beta}$ .

$$\hookrightarrow y(x, w) = w_0 + w_1 x + w_2 x^2$$

a) from prediction

$$\begin{aligned} \mathbb{E}[\hat{t} | x=2] &= y(2, [1, 0.5, -0.1]) = 1 + 0.5(2) + (-0.1)(2^2) \\ &= 1 + 1 - 0.4 = \underline{\underline{1.6}} \end{aligned}$$

b) full probability density

$$\begin{aligned} p(\hat{t} | x, w, \beta) &= \sqrt{\frac{1}{2\pi\sigma^2}} \exp\left(-\frac{(\hat{t} - \mu)^2}{2\sigma^2}\right) \\ &= \sqrt{\frac{\beta}{2\pi}} \exp\left(-\frac{\beta}{2} (\hat{t} - y(x, w))^2\right) \\ &= \underline{\underline{\sqrt{\frac{4}{2\pi}} \exp(-2(\hat{t} - 1.6)^2)}} \end{aligned}$$

# Likelihood function construction

training data:  $\{(x_1, t_1), (x_2, t_2), (x_3, t_3)\} = \{(0, 1.1), (1, 2.9), (2, 4.2)\}$

model:  $y(x, w) = w_0 + w_1 x$

a) individual likelihood terms:

$$p(t_1 | x_1, w, \beta) = \mathcal{N}(1.1 | w_0 + w_1(0), \beta^{-1}) = \mathcal{N}(1.1 | w_0, \beta^{-1})$$

$$p(t_2 | x_2, w, \beta) = \mathcal{N}(2.9 | w_0 + w_1(1), \beta^{-1}) = \mathcal{N}(2.9 | w_0 + w_1, \beta^{-1})$$

$$p(t_3 | x_3, w, \beta) = \mathcal{N}(4.2 | w_0 + w_1(2), \beta^{-1}) = \mathcal{N}(4.2 | w_0 + 2w_1, \beta^{-1})$$

b) Joint likelihood (independence assumption)

$$p(\mathbf{t} | \mathbf{x}, w, \beta) = \prod_{i=1}^N \mathcal{N}(t_i | y(x_i, w), \beta^{-1})$$